

Talstar®

XTRA GRANULAR INSECTICIDE



For use to control pests, as listed, outdoors, including areas surrounding residential, institutional, public, commercial and industrial buildings.

For use on lawns and landscaped areas around residential, institutional, public, commercial and industrial buildings, parks, recreational areas and athletic fields.

Kills fire ant queens within 24 hours.

Kills fire ant mounds within 15 minutes.

EPA Reg. No. 279-9552

EPA Est. 279-IL-1

Active Ingredients:	By Wt.
Zeta-Cypermethrin*	0.05%
Bifenthrin**	0.20%
Other Ingredients:	99.75%
	100.0%

* Cis/trans isomer ratio: Max 75% (±) cis and Min. 25% (±) trans

** Cis isomers 97% minimum; trans isomers 3% maximum.

KEEP OUT OF REACH OF CHILDREN CAUTION

See other panels for additional precautionary information.

FMC

FMC Corporation
1735 Market Street
Agricultural Products Group
Philadelphia PA 19103

Net Contents: 25 Pounds

09/12/12

FIRST AID

If swallowed	• Call poison control center or doctor immediately for treatment advice. • Have person sip a glass of water if able to swallow. • Do not induce vomiting unless told to do so by the poison control center or doctor. • Do not give anything by mouth to an unconscious person..
If in eyes	• Hold eye open and rinse slowly and gently with water for 15-20 minutes. • Remove contact lenses, if present, after the first 5 minutes, then continue rinsing eye. • Call a poison control center or doctor for treatment advice.

HOTLINE NUMBER

Have the product container or label with you when calling a poison control center or doctor, or going for treatment. You may also contact 1-(800)-331-3148 for Emergency Assistance.

NOTE TO PHYSICIAN

This product is a pyrethroid. If large amounts have been ingested, the stomach and intestine should be evacuated. Treatment is symptomatic and supportive. Digestible fats, oils, or alcohol may increase absorption and so should be avoided.

For Information Regarding the Use of this Product Call 1-(800)-321-1FMC (1362).

PRECAUTIONARY STATEMENTS

Hazards to Human (and Domestic Animals) CAUTION

Harmful if swallowed. Causes moderate eye irritation. Avoid contact with eyes or clothing. Prolonged or frequently repeated skin contact may cause allergic reactions in some individuals. **Wear long-sleeved shirt and long pants, socks, shoes and gloves.** Wash thoroughly with soap and water after handling and before eating, drinking, chewing gum, using tobacco or using the toilet.

Environmental Hazards

This pesticide is extremely toxic to fish, aquatic invertebrates, oysters and shrimp. To protect the environment, do not allow pesticide to enter or run off into storm drains, drainage ditches, gutters or surface waters. Applying this product in calm weather when rain is not predicted for the next 24 hours will help to ensure that wind or rain does not blow or wash pesticide off the treatment area. Sweeping any product that lands on a driveway, sidewalk, or street, back onto the treated area of the lawn or garden will help to prevent run off to water bodies or drainage systems. Do not apply when weather conditions favor drift from treated areas. Drift and run-off from treated areas may be hazardous to aquatic organisms in neighboring areas.

STORAGE AND DISPOSAL

Do not contaminate water, food or feed by storage or disposal.

Pesticide Storage

Keep out of reach of children and animals. Store in original containers only. Store in a cool, dry place, preferably in a locked storage area and avoid excess heat. Carefully open containers. After partial use, close tightly.

In case of spill, avoid contact, isolate area and keep out animals and unprotected persons. Confine spills. Call CHEMTREC (transportation and Spills): (800) 424-9300.

To confine spill: Cover to prevent dispersal. Place damaged package in a holding container. Identify contents.

Pesticide Disposal

Wastes resulting from the use of this product may be disposed of on site or at an approved waste disposal facility.

Container Handling

Non-refillable container. Do not reuse or refill this container. Completely empty bag into application equipment. Then offer for recycling, if available, or dispose of empty bag in a sanitary landfill or by incineration or, if allowed by state and local authorities, by burning. If burned, stay out of smoke.

DIRECTIONS FOR USE

It is a violation of Federal Law to use this product in a manner inconsistent with its labeling.

Apply this product directly to the lawn or garden area. Water treated area as directed on this label. Do not water to the point of run-off.

Do not make applications during rain.

Application is prohibited directly into sewers or drains, or to any area like a gutter where drainage to sewers, storm drains, water bodies, or aquatic habitat can occur. Do not allow the product to enter any drain during or after application.

Not for use on plants being grown for sale of other commercial use, or for commercial seed production, or for research purposes.

Not for use on golf courses, sod farms, nurseries, in commercial greenhouses or on grass grown for seed.

For use on plants intended for aesthetic purposes or climate modification and being grown in interior landscapes, ornamental gardens or parks, or lawns or grounds.

Also for use as a perimeter or spot treatment to control pests outdoors, including areas surrounding residential, institutional, public, commercial and industrial buildings.

Do not apply more than 0.25 lb. active ingredient (ai.) per acre (2.3 lbs granules per 1000 sq. ft.) per application on residential use sites (i.e., around private homes, apartment buildings, condominiums, non-agricultural outbuildings, non-commercial greenhouses, pre-schools or day-care facilities).

May be applied at up to 0.50 lb. ai. per acre (4.6 lbs granules per 1000 sq. ft.) per application on non-residential use sites (i.e., around institutional, public, commercial or industrial buildings; parks; recreational areas or athletic fields).

Do not broadcast apply more than 0.5 lb. ai. per acre (4.6 lbs per 1000 sq. ft.) per year.

Lawn Application: Broadcast with suitable application equipment to ensure uniform coverage over the treatment area.

Perimeter Applications: Apply to ground area 5 to 10 feet wide around and adjacent to the structure.

Boundary Applications: Apply to soil areas, ornamental gardens, pavement crevices, fencelines, planter areas, vegetation areas and mulch areas.

Pounds of Talstar® XTRA Granular Insecticide Featuring Verge Granule Technology for Perimeter Applications										
Linear Distance (Feet) Around Structure										
Applic. Rate: Pounds Per 1,000 sq. ft.	Band Width	100	150	200	250	300	400	500	1,000	
2.3	5 feet	1.3	1.9	2.5	3.0	3.6	4.8	5.9	11.7	
2.3	10 feet	3.0	4.2	5.3	6.5	7.6	9.9	12.2	23.7	
3.0	5 feet	1.7	2.5	3.3	3.9	4.7	6.3	7.7	15.2	
3.0	10 feet	3.9	5.4	6.9	8.5	9.9	12.9	15.9	30.9	
3.5	5 feet	2.0	2.9	3.8	4.5	5.4	7.2	8.9	17.5	
3.5	10 feet	4.5	6.3	8.0	9.8	11.4	14.9	18.3	35.5	
4.6	5 feet	2.6	3.8	5.0	6.0	7.2	9.6	11.8	23.4	
4.6	10 feet	6.0	8.4	10.6	13.0	15.2	19.8	24.4	47.4	

Note: The amounts of Talstar® XTRA Granular Insecticide Featuring Verge Granule Technology listed in the above table are based on the total area being treated. This area is determined by multiplying the linear distance by the band width and then adding to that value the area of a circle whose radius is equal to the band width (to account for the area treated at the four corners of a rectangular structure).

APPLICATION RATES

The application rates listed in the following table will provide control of the respective pests under typical conditions.

Pest	Usage per Application		
	Lb Talstar® XTRA Granular Insecticide Featuring Verge per 1000 ft ²	Lb Talstar® XTRA Granular Insecticide Featuring Verge per Acre	Lb Total Active Ingredient per Acre
Annual Bluegrass Weevil (<i>Hyperodes</i>) (adults) Armyworms Billbugs (adults) Black Turfgrass Ataenius (adults) Cutworms Leafhoppers Mealybugs Sod Webworms	1.15 to 2.3 Lbs/1000 sq ft	50 to 100 Lbs per acre	0.125 to 0.25 Lbs AI/Acre
Ants, including carpenter ants (except Pharaoh ants) Imported Fire Ants (adults) Centipedes Crane Flies (larvae) Earwigs Fleas (larvae & adults) Chinch Bugs Millipedes Mole Crickets (nymphs & adults) Pillbugs Scorpions Sowbugs Ticks	2.3 to 4.6 Lbs/1000 sq ft	100 to 200 Lbs per Acre	0.25 to 0.50 Lbs AI/Acre

Resistance Management Recommendations. Talstar® XTRA Granular Insecticide Featuring Verge Granule Technology contains Group 3 Insecticides. Insects with acquired resistance to Group 3 Insecticides may eventually dominate the insect population if Group 3 Insecticides are used repeatedly in successive years as the primary method of control for targeted species. This may result in partial or total loss of control of those species by Talstar® XTRA Granular Insecticide Featuring Verge Granule Technology or other Group 3 Insecticides.

In New York State, this product may NOT be applied to any grass or turf area within 100 feet of a water body (lake, pond, river, stream, wetland, or drainage ditch).

In New York State, do make a single repeat application of Talstar® XTRA Granular Insecticide Featuring Verge Granule Technology if there are signs of renewed insect activity, but not sooner than two weeks after the first application.

SPECIFIC PEST INSTRUCTIONS

For optimum control of particular pests, follow procedures outlined below:

Armyworms, Cutworms, and Sod Webworms: To ensure optimum control, irrigate the treated area with water immediately after application to activate (release from the granule) the insecticide.

Annual Bluegrass Weevil (*Hyperodes*) adults: Applications should be timed to control adult weevils as they leave their overwintering sites and move into grass areas. This movement generally begins when Forsythia is in full bloom and concludes when flowering dogwood is in full bloom. Consult your State Cooperative Extension Service for more specific information regarding application timing.

Billbug adults: Applications should be made when adult billbugs are first observed during April and May. Degree day models have been developed to optimize application timing. Consult your State Cooperative Extension Service for information specific to your region. In temperate regions, spring applications targeting billbug adults will also provide control of over-wintered chinch bugs.

Black Turfgrass Ataenius adults: Applications should be made during May and July to control the first and second generation of black turfgrass Ataenius adults, respectively. The May application should be timed to coincide with the full bloom stage of Vanhoutte spiraea (*Spiraea vanhouttei*) and horse chestnut (*Aesculus hippocastanum*). The July application should be timed to coincide with the blooming of Rose of Sharon (*Hibiscus syriacus*).

Chinch Bugs: Chinch bugs infest the base of grass plants and are often found in the thatch layer. Irrigate the treated area with up to 0.25 inches of water immediately after application to activate (release from the granule) the insecticide. Chinch bugs can be one of the most difficult pests to control in grasses and the higher applications rates may be required to control populations that contain both nymphs and adults during the summer.

Crane Flies: Treatments should be made to control early to mid-season larvae (approximately August – February) as they feed on plant crowns. Treatments made to late-season larvae (approximately March - April) may only provide suppression. Consult your local extensions agent for specific recommendations for your area.

Earwigs: When earwigs are found in mulch, mix granules into the mulch layer. Place a 3-6 foot band of Talstar® XTRA Granular Insecticide

Featuring Verge Granule Technology around the building adjacent to the foundation (perimeter treatment) to stop or limit earwigs from getting indoors. Earwigs are attracted to moisture around homes, particularly when the weather turns hot and dry. For best results, eliminate damp, moist conditions in crawl spaces under houses, around exterior water spigots, air-conditioning units and along house foundations. Also, for best results, reduce lighting around doors, windows and other potential entry sites or use exterior sodium vapor yellow lights which attract fewer insects compared to white, neon, halide, or mercury vapor lights.

Fleas (adults and larvae): Control is provided for up to seven days. Flea larvae develop in the soil of shaded areas that are accessible to pets or other animals. Irrigate the treated area with up to 0.5 inches of water immediately after application to activate (release from the granule) the insecticide.

Imported Fire ants: Control will be optimized by combining broadcast applications that will control foraging workers and newly mated fly-in queens with mound treatments that will control existing colonies. If the soil is not moist, then it is important to irrigate before or soon after application. Broadcast treatments should apply no more than 0.50 lbs ai per acre (4.6 lbs/1000 square feet). Mounds should be treated by applying ½ cup of Talstar® XTRA Granular Insecticide Featuring Verge Granule Technology per mound and then drenching the mound with 1 to 2 gallons of water. The mounds should be treated with sufficient force to break their apex and allow the insecticide solution to flow into the ant tunnels. Treat three feet out around the mound. For best results, apply in cool weather (65 - 80° F) or in early morning or late evening hours. This treatment will control the mound within 15 minutes and kill the queen within 24 hours of application. Application of at least 125 lbs/acre (2.8 lbs granules per 1000 sq. ft.) of Talstar® XTRA Granular Insecticide Featuring Verge Granule Technology will provide four months residual activity for control of foraging Imported Fire Ants and newly mated fly-In Queens.

Mole Cricket adults: Achieving acceptable control of adult mole crickets is difficult because preferred grass areas are subject to continuous invasion during the early spring by this extremely active stage. Applications should be made as late in the day as possible and should be watered in with up to 0.5 inches of water immediately after treatment. If the soil is not moist, then it is important to irrigate before application to bring the mole crickets closer to the soil surface where contact with the insecticide will be maximized. Grass areas that receive pressure from adult mole crickets should be treated at peak egg hatch to ensure optimum control of subsequent nymph populations (see below).

Mole Cricket nymphs: Grass areas that received intense adult mole cricket pressure in the spring should be treated immediately prior to peak egg hatch. Optimal control is achieved at this time because young nymphs are more susceptible to insecticides and they are located near the soil surface where the insecticide is most concentrated. Control of larger, more damaging, nymphs later in the year may require both higher application rates and more frequent applications to maintain acceptable control. Applications should be made as late in the day as possible and should be watered in with up to 0.5 inches of water immediately after treatment. If the soil is not moist, then it is important to irrigate before application to bring the mole crickets closer to the soil surface where contact with the insecticide will be maximized.

Scorpions: To ensure optimum control, treat the building perimeter at dusk to prevent outdoor scorpions from entering the building. Scorpions reside hidden in cracks and voids during the day and are active at night at temperatures above 77°F/25°C. Resting areas can be identified at night using a blacklight (UV bulb) as scorpions will fluoresce. These areas should also be treated. Prior to treatment, to increase treatment efficacy, remove trash, debris, or firewood that scorpions may use as resting sites.

Ticks (Including ticks that may transmit Lyme Disease and Rocky Mountain Spotted Fever): Do not make spot applications. Treat the entire area where exposure to ticks may occur. Use higher application rates when treating areas with dense ground cover or heavy leaf litter. Ticks may be reintroduced from surrounding areas on host animals. Do not allow public use of treated areas during application.

Deer ticks (*Ixodes sp.*) have a complicated life cycle that ranges over a two year period and involves four life stages. Applications should be made in mid to late-spring to control larvae and nymphs that reside in the soil and leaf litter.

American dog ticks may be a considerable nuisance in suburban settings, particularly where homes are built on land that was previously field or forest. These ticks commonly congregate along paths or roadways where humans are likely to be encountered. Applications should be made as necessary from mid-spring to early fall to control American dog tick larvae, nymphs and adults.

SPOT TREATMENT APPLICATIONS

Mound Ant Treatment, including Fire Ants: Apply 1/2 cup of Talstar® XTRA Granular Insecticide Featuring Verge Granule Technology per mound and then drench the mound with 1 to 2 gallons of water. The mound should be drenched with sufficient force to break its apex and allow the insecticide to flow into the ant tunnels. Treat three feet out around the mound. For best results, apply in cool weather (65 to 80°F) or in early morning or late evening hours. Treat new mounds as they appear. Talstar® XTRA Granular Insecticide Featuring Verge Granule Technology will provide control of ant mounds within four hours. Food

utensils such as teaspoons and measuring cups should not be used for food purposes after use with insecticide.

INDUSTRIAL NON-AGRICULTURAL APPLICATIONS

(Including Industrial Sites, Parks, Ball Fields and Cemeteries)

Broadcast and Spot Applications: For the control of actively foraging ants (including Imported Fire Ants) broadcast uniformly 2.3 to 4.6 lbs Talstar® XTRA Granular Insecticide Featuring Verge Granule Technology per 1000 sq. ft.

Pad Mounted Electrical Sites: To reduce or prevent ant mound building activity on electrical pads, uniformly distribute 1.0 to 1.5 Tablespoons of Talstar® XTRA Granular Insecticide Featuring Verge Granule Technology per 4 sq. ft. of mounting pad. Where existing transformers are completely or partially filled with soil excavated by resident ants, remove any excess fill and level the remaining soil prior to application. **Food utensils such as tablespoons must not be used for food purposes after use in measuring pesticides.**

Note: Electrical transformers appear to be more attractive to ants during the cooler months. For best results treat in the Fall or early Spring.

SPREADER EQUIPMENT SETTINGS

Broadcast application should be performed with a calibrated commercial broadcast spreader to evenly distribute the granules. Approximate settings for several models are provided in the table below.

Talstar® XTRA Granular Insecticide Featuring Verge™ Granule Technology - Suggested Spreader Setting Guide									
Application Rate lbs/1,000 sq ft	Lesco® Broadcast - JD Rotary (TY 2332)	Lesco® Broadcast - Old	Lesco Spreader Model 091186	Scotts® HandyGreen II	Scotts® SpeedyGreen 2000	Andersons Accu-Pro 2000, Cone Setting: 8, 3 MPH	Earthway - E-V-N-Spread Drop	Earthway - E-V-N-SPRED - Rotary, 3.0 MPH	Earthway - E-V-N-SPRED - Rotary (2000), 3.5 MPH
1.15 lbs	2.8	F	8.3	4	2 3/4	6.6 I	12.4	13.4	4.5
2.3 lbs	4	G	10.4	4*	5 1/4	7.5 J	12.6	14.5	8
4.6 lbs	5.8	I	14.6	4**	9	9.2 K	13.1	16.8	14
Width Ft.	12	10	8	10	10		1.75		11
*Use this setting & make 2 passes over treatment area to achieve 2.3 lbs/1,000 sq ft									
**Use this setting & make 4 passes over treatment area to achieve 4.6 lbs/1,000 sq ft									

It is recommended that you calibrate your own spreader periodically.

RESISTANCE MANAGEMENT: Some insects are known to develop resistance to products used repeatedly for control. Because the development of resistance cannot be predicted, the use of this product should conform to resistance management strategies established for the use area. Consult your local or state pest management authorities for details.

If resistance to this product develops in your area, this product, or other products with a similar mode of action, may not provide adequate control. If poor performance cannot be attributed to improper application or extreme weather conditions, a resistant strain of insect may be present. If you experience difficulty with control and suspect that resistance is a reasonable cause, immediately consult your local company representative or pest management advisor for the best alternative method of control for your area.

Conditions of Sale and Limitation of Warranty and Liability:

Notice: Read the entire Directions for Use and Conditions of Sale and Limitation of Warranty and Liability before buying or using this product. If the terms are not acceptable, return the product at once, unopened, and the purchase price will be refunded.

The Directions for Use of this product must be followed carefully. It is impossible to eliminate all risks inherently associated with the use of this product. Crop injury, ineffectiveness, or other unintended consequences may result because of such factors as manner of use or application, weather or crop conditions beyond the control of FMC or Seller. All such risks shall be assumed by Buyer and User, and, to the extent permitted by applicable law, Buyer and User agree to hold FMC and Seller harmless for any claims relating to such factors.

Seller warrants that this product conforms to the chemical description on the label and is reasonably fit for the purposes stated on the Directions for Use when used in accordance with the directions under normal conditions of use. TO THE EXTENT PERMITTED BY APPLICABLE LAW, FMC MAKES NO WARRANTIES OF MERCHANTABILITY OR OF FITNESS FOR A PARTICULAR PURPOSE, NOR ANY OTHER EXPRESS OR IMPLIED WARRANTIES WITH RESPECT TO THE SELECTION, PURCHASE, OR USE OF

THIS PRODUCT. Any warranties, express or implied, having been made are inapplicable if this product has been used contrary to label instructions, or under abnormal conditions, or under conditions not reasonably foreseeable to (or beyond the control of) seller or FMC, and, to the extent permitted by applicable law, buyer assumes the risk of any such use.

To the extent permitted by applicable law, FMC or seller shall not be liable for any incidental, consequential or special damages resulting from the use or handling of this product. TO THE EXTENT PERMITTED BY APPLICABLE LAW, THE EXCLUSIVE REMEDY OF THE USER OR BUYER, AND THE EXCLUSIVE LIABILITY OF FMC AND SELLER FOR ANY AND ALL CLAIMS. LOSSES, INJURIES OR DAMAGES (INCLUDING CLAIMS BASED ON BREACH OF WARRANTY, CONTRACT, NEGLIGENCE, TORT, STRICT LIABILITY OR OTHERWISE) RESULTING FROM THE USE OR HANDLING OF THIS PRODUCT, SHALL BE THE RETURN OF THE PURCHASE PRICE OF THE PRODUCT OR, AT THE ELECTION OF FMC OR SELLER, THE REPLACEMENT OF THE PRODUCT.

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